

PHY 311 — Modern Physics

Fall 2026 · Klamath Falls / Portland-Metro (Zoom-linked) · TuTh, 3:00–4:20 p.m.

Instructors	David Johnston	Jesse Kinder
Office hours	TBD	Mon/Wed 3:30–5:30 p.m., Tue/Thu 4:30–5:30 p.m.

Required Text: *Six Ideas That Shaped Physics*, Units R (Relativity) and Q (Quantum) — Moore

Supplement: *Exploring Black Holes* (Taylor & Wheeler) — selected sections for the GR unit

Recommended: *Spacetime Physics* (Taylor & Wheeler) — relativity problem source

Grades and assignments live on Canvas. Everything else is here.

Learning Objectives

By the end of this course, you should be able to:

- **Think like a physicist.** Apply relativity and quantum mechanics to explain, analyze, and predict real phenomena.
- **Think critically.** Identify where relativity and quantum mechanics defy common sense, and trust evidence over instinct. Nature is always right.
- **Solve problems.** Apply problem-solving heuristics in relativity and quantum mechanics. Communicate your reasoning clearly, on paper and at the board.
- **Connect theory to experiment and application.** Describe key experiments that shaped modern physics. Explain technologies — MRI, GPS, semiconductors — that depend on them today.

Format

This course is co-taught by David and Jesse.

Physics education research is consistent on one point — you learn physics by doing it, not by watching someone talk about it — so most of our time together is discussion, problem solving, and chalk talks at the board, not lecture.

Come prepared: read the assigned material *before* class, and bring pencil, a calculator, and your laptop or tablet.

We will use a leapfrog method as we work through the text.

- Read the chapter and attempt a few practice problems before class.

- Ask questions. Discuss and apply the new concepts during class.
- Attempt challenging problems after class.
- Repeat.

Grading

There are no exams in this course. Your grade comes from problem sets, chalk talks, and two projects.

Component	Percentage
Problem Sets	50
Chalk Talks	15
Midterm Project	15
Final Project	20

Problem Sets include practice problems due before class each day and challenge problems due each week. Solutions to the challenge problems will be available for self-assessment and corrections. Most of your grade on problem sets (30 points) comes from completion. The remaining points will be assigned at the end of the term by looking at the overall quality of your body of work: effort, clarity, accuracy, critical thinking, and reflection.

Chalk talks. Much of class time is discussion — of problems, of your questions. You'll occasionally share your own work informally at a chalkboard. Regular attendance and visible, steady effort earns full credit here. You don't need the right answer to get credit for a chalk talk. There's a difference between being confused and being unprepared, though.

Midterm project. Research a key experiment in the development of modern physics and present it as we discuss the topic in class. These will be spread throughout the term.

Final project. Research a technology application of relativity or quantum physics. Prepare a written report and peer-review classmates' reports.

Academic Integrity

Work with others, use outside resources, learn from peers and tutors — but submit your own original work. What you turn in should show your own reasoning, your own calculations, your own conclusions.

Generative AI is useful for composing text — use it. But don't ask it to do your assignments for you: you do the thinking, AI can help with the writing. If you use it, say so on the assignment and share a link to the session. Undisclosed AI use counts as academic dishonesty here, the same as any other uncredited source.

Plagiarism — copying someone else's work, or letting your own be copied — earns zero credit for the copy, and possibly for the original too. Egregious violations (sabotage, forged documents, computer fraud) go to the Student Hearing Commission. Review Oregon Tech's [Student Academic Integrity Policy](#) if you're not already familiar with it.

Accessibility

If you need accommodation because of a disability, contact us at the start of the term and reach out to [Disability Services](#) — we rely on them to coordinate accommodations.

University Policies

Oregon Tech’s [Commission on College Teaching syllabus statements](#) and the “Campus-Wide Syllabus Information” document on Canvas are part of this syllabus. Where they conflict with this document, this document takes precedence.

Tentative Schedule

- Leapfrog through Unit R (Relativity, ~3 weeks)
- Quick survey of general relativity and the experiments that ground the physics (selections from *EBH*, ~2 weeks)
- Leapfrog through Unit Q (Quantum, ~5 weeks)

Experiment names below are short on purpose — the full story behind each one (context, why it’s the right test, Nobel Prizes where they genuinely apply) comes out in class.

Date	Reading	Topic	Experiments
Oct 1	—	Course intro; Moore’s notation; start R1	Michelson–Morley
Oct 6	R1 & R2	Coordinate time	Hafele–Keating
Oct 8	R3	Spacetime interval	Kennedy–Thorndike
Oct 13	R4 & R5	Proper time; coordinate transformations	Rossi–Hall (muons)
Oct 15	R6	Lorentz contraction	<i>(discussion day — revisit Rossi–Hall)</i>
Oct 20	R7 & RB	Cosmic speed limit; Doppler effect	Bertozzi · Ives–Stillwell
Oct 22	R8 & R9	Four-momentum, and its conservation	Anderson (positron) · Compton scattering
Oct 27	EBH 1 & 2	SR review → the curvature bridge to GR	<i>(bridge day — no new experiment)</i>
Oct 29	EBH 3 & 4	Schwarzschild map; GPS	Pound–Rebka
Nov 3	EBH 6, 7 & 16	Falling in; gravitational waves	LIGO · Cygnus X-1 · EHT
Nov 5	EBH 10 & 11	GR’s classic confirmations	Eddington · Mercury’s perihelion · 2020 Nobel Prize
Nov 10	<i>review</i> Q1, Q2 — <i>new</i> Q3	Interference and diffraction	<i>(bridge/review day)</i>
Nov 12	Q4 & Q5	Photons; matter waves	Compton / Bothe–Geiger · Davisson–Germer
Nov 17	Q6 & Q7	Spin; rules of QM	Stern–Gerlach · Franck–Hertz
Nov 19	QA & Q8	Complex numbers; quantum weirdness	Bell tests (2022 Nobel Prize)
Nov 24	Q9 & Q10	Wavefunction; simple quantum models	Jönsson (electron double-slit)
—	—	<i>No class — Thanksgiving</i>	
Dec 1	Q11 & Q12	Spectra; the Schrödinger equation	Lamb shift
Dec 3	Q13 & Q14	Nuclei; nuclear stability — background for Jaime’s talk	Rutherford · Cowan–Reines
Dec 8	—	Guest speaker: Jaime Evers , medical physics & nuclear power	
Dec 10	Q15	Nuclear technology — closes the term, follows up on Jaime’s talk	